

CUSUM Change-Point Detection Analysis

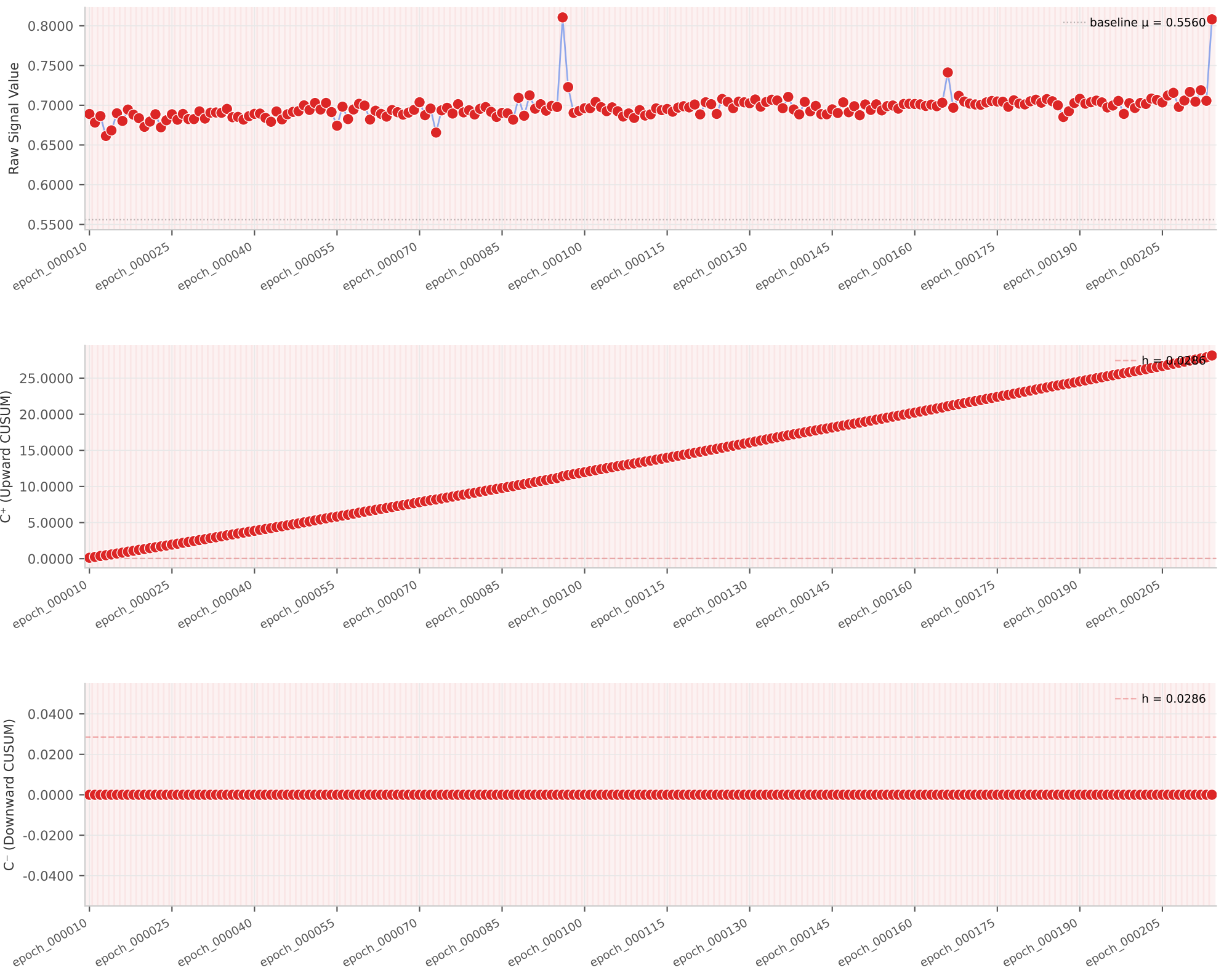
Cumulative Sum Control Chart | JS Divergence + New IP Ratio + Top-K Source

Run Parameters		Baseline CUSUM Parameters			Detection Summary (Any Signal)	
Dataset	epochs_5.0_2.0_cusum.json	Combined JS	$\mu=0.5560$	$\sigma=0.0071$	$k=0.0036$ $h=0.0286$	205
Timestamp	2026-02-23T20:12:41.301642	New IP Ratio	$\mu=0.0000$	$\sigma=0.0000$	$k=0.0000$ $h=0.0000$	0
Method	CUSUM Change-Point (JS + New IP + TopK)		$\mu=0.0622$	$\sigma=0.0031$	$k=0.0015$ $h=0.0123$	0
Total epochs	215	False Negatives (FN)				0
Baseline epochs	10 (5.0%)	Precision				1.000
Attack epochs	205	Recall				1.000
Malicious traffic	2.0%	F1 Score				1.000
CUSUM k-factor	0.5	Detection Rate				100.0%
CUSUM h-factor	4.0					
Mode	FULL					

Combined JS Divergence

CUSUM Signal Analysis | malicious = 2.0% | baseline = 5.0%

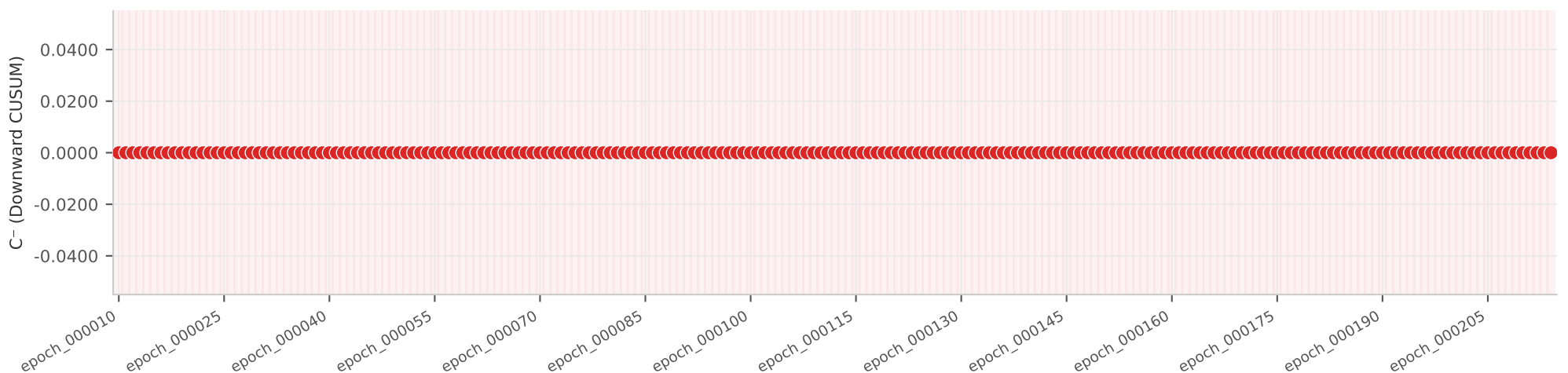
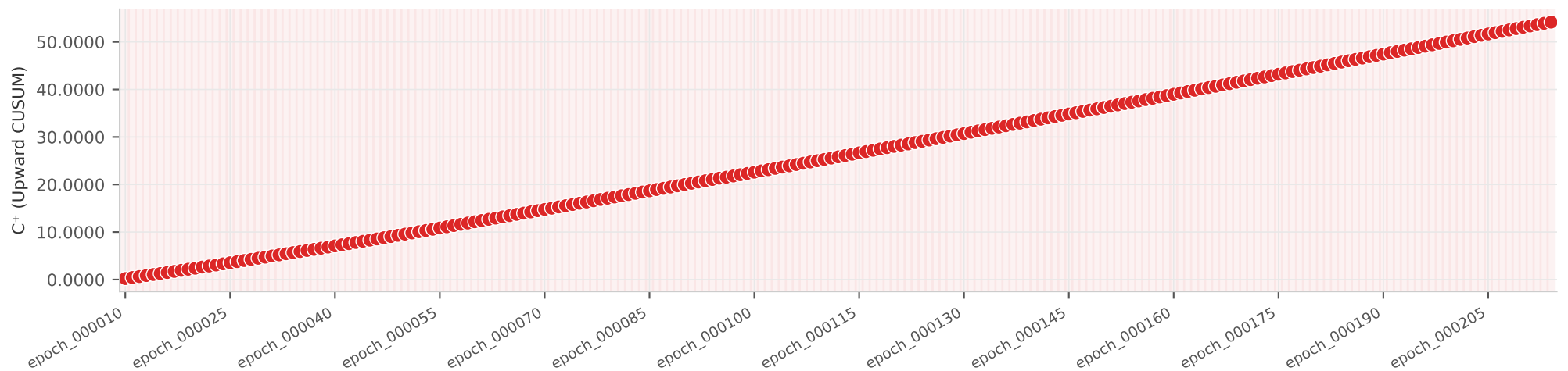
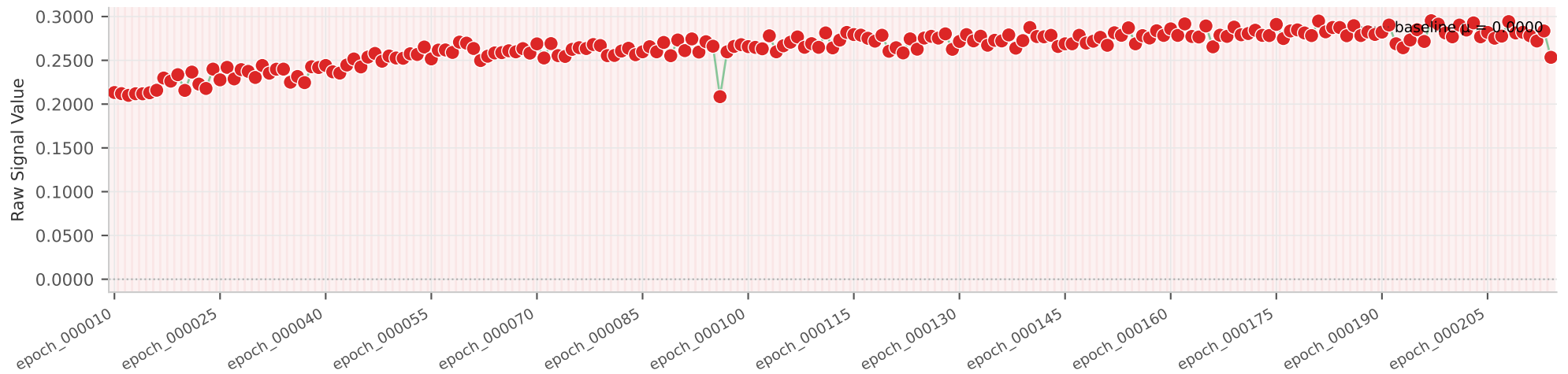
- CUSUM alarm fired
- Attacked epoch



New Source IP Ratio

CUSUM Signal Analysis | malicious = 2.0% | baseline = 5.0%

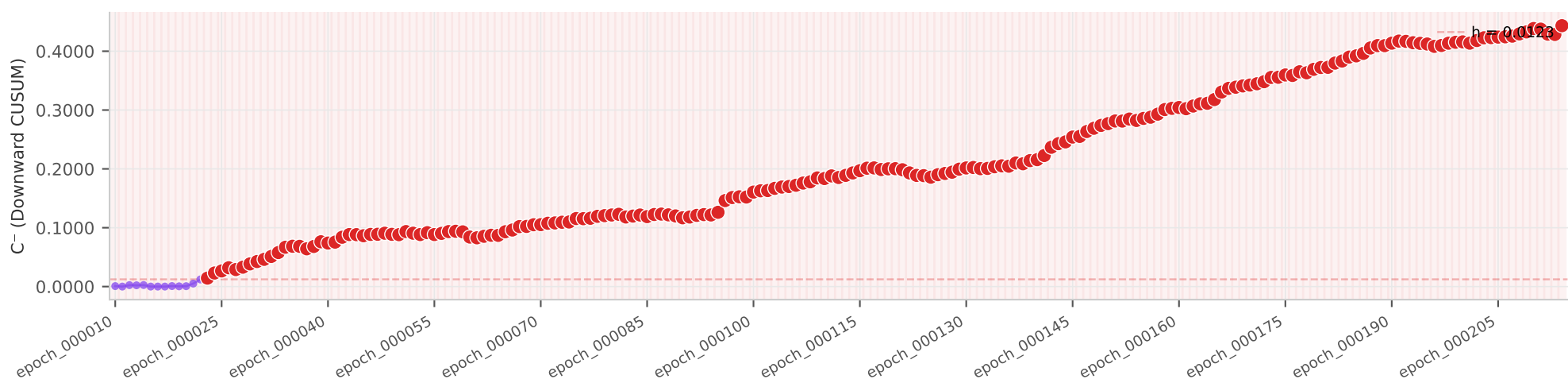
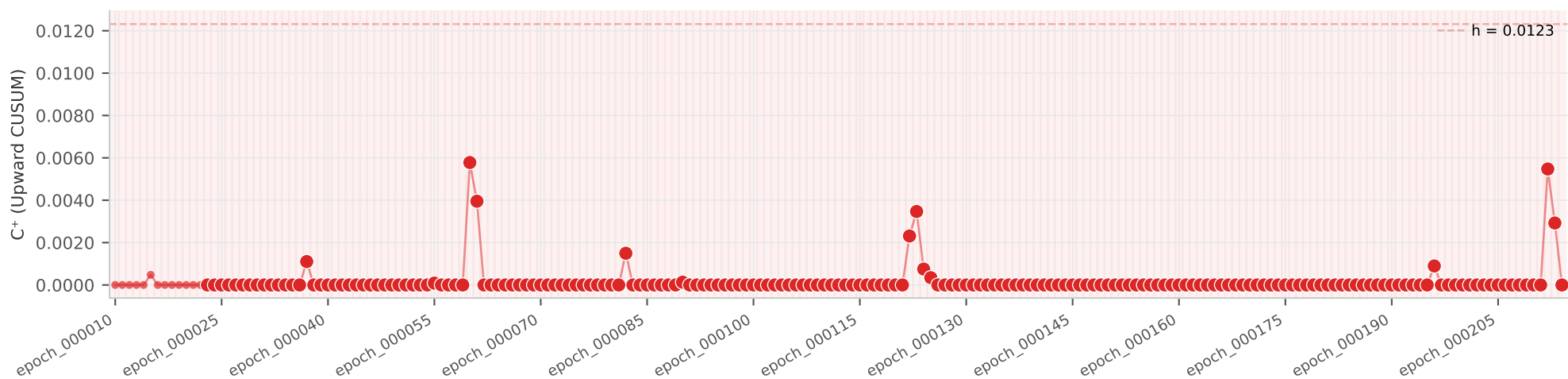
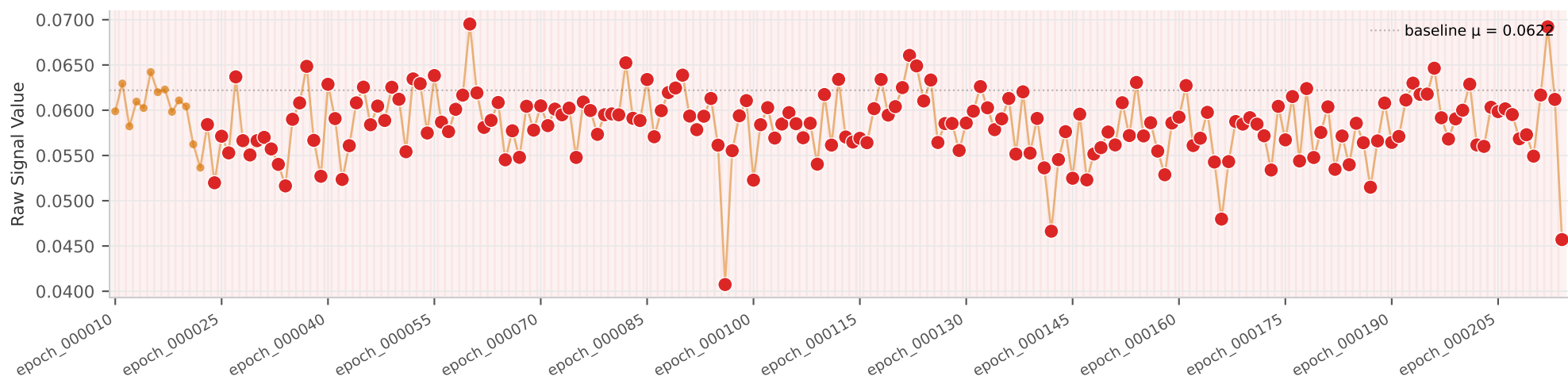
- CUSUM alarm fired
- Attacked epoch



Top-5 Source Concentration

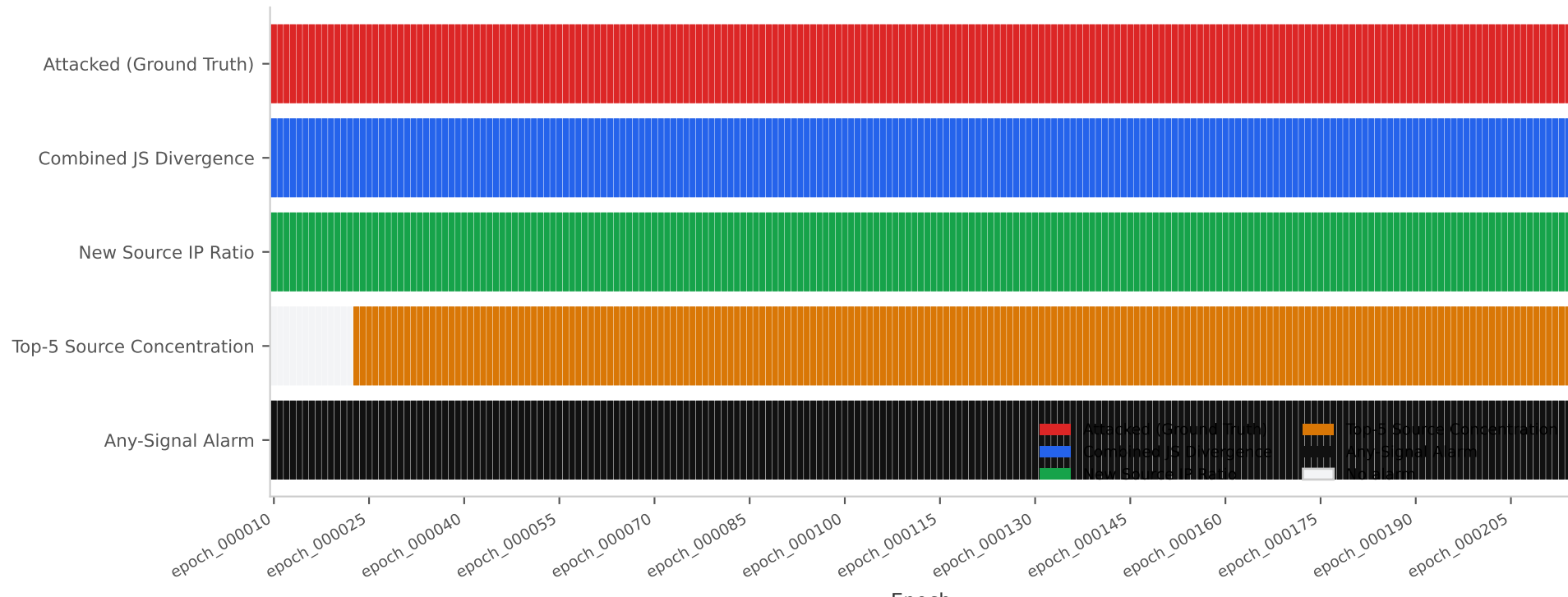
CUSUM Signal Analysis | malicious = 2.0% | baseline = 5.0%

- CUSUM alarm fired
- Attacked epoch



CUSUM Alarm Timeline

Filled cell = alarm fired for that epoch × signal



Detection Performance Metrics

Dataset: epochs | Malicious: 2.0% | Baseline: 5.0%

Metric	Value	Description
True Positives	205	Attacked epochs correctly alarmed
False Positives	0	Clean epochs incorrectly alarmed
True Negatives	0	Clean epochs correctly not alarmed
False Negatives	0	Attacked epochs missed
Precision	1.0000	$TP / (TP + FP)$
Recall	1.0000	$TP / (TP + FN)$
F1 Score	1.0000	Harmonic mean of Precision & Recall
Detection Rate	100.00%	Fraction of attacked epochs detected
False Alarm Rate	0.00%	Fraction of clean epochs with alarm